

RELATIONAL ALGEBRA (RA)

University Database Schema:

Student (SID, SName, address, dept, totalCredits) -- PK: SID

StudentCourse (SID, CCode, Grade) -- PK: SID, CCode

-- FK: SID references Student(SID)

-- FK: CCode references Course(CCode)

ST (SID, assignedCourse) -- PK: SID

-- FK: SID references Student(SID)

-- FK: assignedCourse references Course(CCode)

Faculty (FIN, FName, Desk#) -- PK: FIN

Course (CCode, CName) -- PK: CCode

FacultyCourses (FIN, CCode, Section) -- PK: FIN, CCode, Section

-- FK: FIN references Faculty(FIN)

-- FK: CCode references Course(CCode)

Write RA to solve the following problems

Note: There are multiple ways to solve the problems and some of these RA queries may not exactly match your solutions. For more practice, you are encouraged to try solving the problems in multiple ways.

1. Retrieve all information of Students who are from 'CS' or 'MNS' dept.
2. Retrieve all information of Students who are not from 'CS' or 'MNS' dept.
3. Find SName, address, dept for students whose totalCredits is between 50 to 70.
4. Rename table "Faculty" to "Teacher".
5. Rename SID to STID and assignedCourse to Course for the ST table.
6. Find SID, SName, address, dept for ST's who are assigned to 'CSE111'.
7. Find the SID, grade for Students who are from 'CSE' dept.
8. Find the CCode of all courses taught by faculty with the FName "Mr. Tom".
9. List all students name, dept and assignedCourse.
10. Find the names of faculty and the course names they teach, where the section is "01".
11. List the names of students and the course names they are taking, where the student is from the "CSE" department.
12. Find the names of faculties who are taking the course "Database Systems".
13. Find the names of students who have received the grade 'A' for the course named 'Database Systems'.
14. Find the names of students who are not an ST.

Write RA to convert the following SQL queries.

15. SELECT distinct SName, dept FROM Student WHERE dept = 'EEE';
16. SELECT * FROM StudentCourse WHERE Grade = 'A';
17. SELECT distinct FIN, FName FROM Faculty, FacultyCourses, Course
WHERE Faculty.FIN = FacultyCourses.FIN
AND FacultyCourses.CCode = Course.CCode
AND Course.CName = 'Automata and Computability';
18. SELECT distinct SID, SName, CName FROM Student, ST, Course
WHERE Student.SID = ST.SID
AND ST.assignedCourse = Course.CCode;

[Optional: For more practice,

(1) write SQL to solve the problems, inspect and compare your SQL and RA;

(2) try to solve as many problems & alternative ways as possible]

Solution:

1. Retrieve all information of Students who are from 'CS' or 'MNS' dept.
 $\sigma_{\text{dept}='CS' \text{ or } \text{dept}='MNS'}}(\text{Student})$
 2. Retrieve all information of Students who are not from 'CS' or 'MNS' dept.
 $\sigma_{\text{dept} \neq 'CS' \text{ or } \text{dept} \neq 'MNS'}}(\text{Student})$
 3. Find SName, address, dept for students whose totalCredits is between 50 to 70.
 $\pi_{\text{SName, address, dept}}(\sigma_{\text{totalCredits} \geq 50 \text{ and } \text{totalCredits} \leq 70}(\text{Student}))$
 4. Rename table "Faculty" to "Teacher".
 $\rho_{\text{Teacher}}(\text{Faculty})$
 5. Rename SID to STID and assignedCourse to Course for the ST table.
 $\rho_{\text{STID, Course}}(\text{ST})$
 6. Find SID, SName, address, dept for ST's who are assigned to 'CSE111'.
 $\pi_{\text{SID, SName, address, dept}}(\sigma_{\text{assignedCourse}='CSE111'}(\text{Student} \bowtie_{\text{SID}=\text{STID}} \text{ST}))$
 7. Find the SID, grade for Students who are from 'CSE' dept
 $\pi_{\text{SID, Grade}}(\sigma_{\text{dept}='CSE'}(\text{Student} \bowtie_{\text{SID}=\text{SID}} \text{StudentCourse}))$
 8. Find the CCode of all courses taught by faculty with the FName "Mr. Tom".
 $\pi_{\text{CCode}}(\sigma_{\text{FName}='Mr. Tom'}(\text{Faculty} \bowtie_{\text{FIN}=\text{FIN}} \text{FacultyCourses}))$
 9. List all students name, dept and assignedCourse.
 $\sigma_{\text{SName, dept, assignedCourse}}(\text{Student} \bowtie_{\text{SID}=\text{STID}} \text{ST})$
 10. Find the names of faculty and the course names they teach, where the section is "01".
 $\pi_{\text{FName, CName}}(\sigma_{\text{Section}='01'}((\text{Faculty} \bowtie_{\text{FIN}=\text{FIN}} \text{FacultyCourses}) \bowtie_{\text{CCode}=\text{CCode}} \text{Course}))$
 11. List the names of students and the course names they are taking, where the student is from the "CSE" department.
 $\pi_{\text{SName, CName}}(\sigma_{\text{dept}='CSE'}((\text{Student} \bowtie_{\text{SID}=\text{SID}} \text{StudentCourse}) \bowtie_{\text{CCode}=\text{CCode}} \text{Course}))$
 12. Find the names of faculties who are taking the course "Database Systems".
 $\pi_{\text{FName}}(\sigma_{\text{CName}='Database Systems'}(\text{Faculty} \bowtie_{\text{FIN}=\text{FIN}} \text{FacultyCourses}) \bowtie_{\text{CCode}=\text{CCode}} \text{Course})$
 13. Find the names of students who have received the grade 'A' for the course named 'Database Systems'.
 $R \leftarrow (\text{Student} \bowtie_{\text{SID}=\text{SID}} \text{StudentCourse}) \bowtie_{\text{CCode}=\text{CCode}} \text{Course}$
 $\pi_{\text{SName}}(\sigma_{\text{Grade}='A' \text{ and } \text{CName}='Database Systems'}(R))$
 14. Find the names of students who are not an ST.
 $\pi_{\text{SName}}(\text{Student}) - \pi_{\text{SName}}(\text{Student} \bowtie_{\text{SID}=\text{STID}} \text{ST})$
 15. SELECT distinct SName, dept FROM Student WHERE dept = 'EEE';
 $\pi_{\text{SName, dept}}(\sigma_{\text{dept}='EEE'}(\text{Student}))$
 16. SELECT * FROM StudentCourse WHERE Grade = 'A';
 $\sigma_{\text{Grade}='A'}(\text{StudentCourses})$
 17. SELECT distinct FIN, FName FROM Faculty, FacultyCourses, Course
WHERE Faculty.FIN = FacultyCourses.FIN
AND FacultyCourses.CCode = Course.CCode
AND Course.CName = 'Automata and Computability';
 $\pi_{\text{FIN, FName}}(\sigma_{\text{CName}='Database Systems'}((\text{Faculty} \bowtie_{\text{FIN}=\text{FIN}} \text{FacultyCourses}) \bowtie_{\text{CCode}=\text{CCode}} \text{Course}))$
 18. SELECT distinct SID, SName, CName FROM Student, ST, Course
WHERE Student.SID = ST.SID
AND ST.assignedCourse = Course.CCode;
 $\pi_{\text{SID, SName, CName}}((\text{Student} \bowtie_{\text{SID}=\text{STID}} \text{ST}) \bowtie_{\text{AssignedCourse}=\text{CCode}} \text{Course})$
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